Tianyi Xiong

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Education

- **University of Maryland, College Park, USA** — Aug 2023 - Present
  Ph.D. in Computer Science
  - Research Interests: Post-training for multimodal foundation models, reward and critic modeling, LLM/VLM-based agents, multimodal evaluation, and recursive self-improvement.
  - M.S. in Computer Science, Aug 2023 - Aug 2026.
- **Tsinghua University, China** — Aug 2019 - Jun 2023
  B.Eng. in Computer Science and Technology (with honors, minor in Statistics)

Selected Publications and Preprints

For a complete list, please visit my Google Scholar profile.

1. **Tianyi Xiong**, Zhengyuan Yang, Xiaofei Wang, Chung-Ching Lin, Ruichun Ma, Kevin Lin, Zhendong Wang, Linjie Li, Chenxi Liu, Ruibo Chen, Ramani Duraiswami, Heng Huang, Lijuan Wang. "Rubrics as Visual-Repair Context for Self-Evolving UI-to-Code Generation". (NeurIPS'26 LCFM Workshop) [paper] [project]
2. Chenxi Liu, **Tianyi Xiong**, Yanshuo Chen, Heng Huang. "Grounding Multimodal Reasoning with Evidence-Ablated Negatives". In *Advances in Neural Information Processing Systems*, 2026. (NeurIPS'26)
3. Tong Zheng, Haolin Liu, Chengsong Huang, Huiwen Bao, Sheng Zhang, Rui Liu, Runpeng Dai, Ruibo Chen, Chenxi Liu, **Tianyi Xiong**, Xidong Wu, Hongming Zhang, Heng Huang. "LLMs Improving LLMs: Agentic Discovery for Test-Time Scaling". In *Advances in Neural Information Processing Systems*, 2026. (NeurIPS'26) [paper] [project]
4. Tianfu Wang, Mingyang Xie, Haoming Cai, **Tianyi Xiong**, Xiyao Wang, Dongdong Fu, Guan-Ming Su, Paola Cascante-Bonilla, Christopher A. Metzler. "All-Weather VLM: Enhancing Vision Language Models' Robustness Under Adverse Imaging Conditions." In *The Third Annual Conference on Language Modeling*, 2026. (COLM'26) [paper]
5. **Tianyi Xiong**, Shihao Wang, Guilin Liu, Yi Dong, Ming Li, Heng Huang, Jan Kautz, Zhiding Yu. "PhyCritic: Multimodal Critic Models for Physical AI". In *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2026. (CVPR'26) [paper] [project]
6. **Tianyi Xiong**, Yi Ge, Ming Li, Zuolong Zhang, Pranav Kulkarni, Kaishen Wang, Qi He, Zeying Zhu, Chenxi Liu, Ruibo Chen, Tong Zheng, Yanshuo Chen, Xiyao Wang, Renrui Zhang, Wenhu Chen, Heng Huang. "Multi-Crit: Benchmarking Multimodal Judges on Pluralistic Criteria-Following". In *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2026. (CVPR'26 Highlight) [paper] [project]
7. Xiyao Wang, Chunyuan Li, Jianwei Yang, Kai Zhang, Bo Liu, **Tianyi Xiong**, Furong Huang. "LLaVA-Critic-R1: Your Critic Model is Secretly a Strong Policy Model". (arXiv'25) [paper]
8. **Tianyi Xiong**, Xiyao Wang, Dong Guo, Qinghao Ye, Haoqi Fan, Quanquan Gu, Heng Huang, Chunyuan Li. "LLaVA-Critic: Learning to Evaluate Multimodal Models". In *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025. (CVPR'25) [paper] [project]
9. **Tianyi Xiong***, Jiayi Wu*, Botao He, Cornelia Fermuller, Yiannis Aloimonos, Heng Huang, Christopher A. Metzler. "Event3DGS: Event-Based 3D Gaussian Splatting for High-Speed Robot Egomotion" In *8th Annual Conference on Robot Learning*, 2024. (CoRL'24) [paper] [project]
10. Xin Xu*, **Tianyi Xiong***, Zheng Ding, Zhuowen Tu. "MasQCLIP for Open-Vocabulary Universal Image Segmentation." In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023. (ICCV'23) [paper] [project]

Experience

- *Research Intern* | **Meta Superintelligence Labs** — Jun 2026 - Dec 2026 (expected)
  Mentors: Sean Sun, Annie Dong — *Menlo Park, CA*
  - Developing a code-instructed agentic data generation pipeline for multimodal image preference and reward modeling.
- *Research Intern* | **Microsoft Research** — Dec 2025 - May 2026
  Mentors: Zhengyuan Yang, Lijuan Wang — *Remote*
  - Developed a rubric-based self-evolving framework for stable and consistent recursive self-improvement in UI-to-code generation. [RubSE, NeurIPS'26 LCFM Workshop]
- *Research Intern* | **NVIDIA Research** — Feb 2025 - Aug 2025
  Mentors: Zhiding Yu, Yi Dong, Guilin Liu — *Santa Clara, CA*

- Developed a **VLM-based critic for physical AI** using a two-stage RL pipeline and introduced **self-referential critic fine-tuning**—a solve-then-judge approach that grounds evaluations in the model's internally generated predictions and reasoning, yielding stronger judgment consistency and improved physical reasoning within a single model. [PhyCritic, CVPR'26]
- Member of the NVIDIA Eagle team, engaged in the development of next-generation foundation vision-language models (VLMs), and contributed to the NVIDIA GR00T N1.6 project.

• *Research Intern* | **ByteDance Seed** Jun 2024 - Nov 2024
Mentor: Chunyuan Li *Bellevue, WA*

- Introduced the **first open-source large multimodal model as a generalist evaluator** to assess model performance across diverse multimodal tasks. Demonstrated its effectiveness in: (i) Judging model responses with image-grounded reasons; (ii) Providing reward signals for preference alignment. [LLaVA-Critic, CVPR'25]

• *Research Intern* | **Microsoft Research Asia** Jan 2023 - Apr 2023
Mentor: Zhirong Wu and Steve Lin *Beijing, China*

- Utilized **self-supervised ViTs** for generating **instance-level mask proposals** for images and videos, improved existing graph-based segmentation algorithms.

• *Research Intern* | **University of California, San Diego** Jul 2022 - Nov 2022
Advisor: Prof. Zhuowen Tu *Remote*

- **Unified open-vocabulary image segmentation** into a two-stage pipeline comprising a class-agnostic mask generator and a CLIP-based mask classifier. Proposed **progressive training** for generating unseen masks, developed an effective and efficient strategy of **fine-tuning CLIP** for adaptation. [MasQCLIP, ICCV'23]

Services

- Reviewer of MM'24, ICLR'25'26, CVPR'25'26, NeurIPS'25'26, ICML'26, TMLR
- Guest Lecturer and Head Teaching Assistant, CMSC848G (2023–2025)

Selected Awards

- Dean's Fellowship, University of Maryland, 2023-2024
- China National Scholarship, Sep 2020
- 35th Chinese Physics Olympiad (Final), Second Award, Oct 2018

Skills

- **Programming & Technical Skills**: Python, C/C++, PyTorch, Hugging Face Transformers; LLM/VLM post-training, reinforcement learning, reward/critic modeling, multimodal evaluation and benchmarking, and agentic systems.
- **Sports**: soccer school team, Champion of the 2020 Beijing University Soccer League; track and field school team
- **Volunteer**: 600+ hours; Docent, Soong Ching Ling Residence (2016-19); Doping Control Team, Beijing 2022 Olympics